AIRLINE TICKET PRICES VS FUEL COSTS ANALYSIS

A Data-Driven Study on the Impact of Fuel Price Volatility on Airline Fare Dynamics

Sector - Aviation Industry

FACULTY - PROF. ARCHIT RAJ

Group 18

https://github.com/connectwithvanshika/C_G18_AirlineTicketPricesVsFuelCostsAnalysis

https://public.tableau.com/app/profile/syed.darain.qamar7769/viz/AirlinePricesAnalysis/ExecutiveOverview?publish=yes

TEAM MEMBERS

Vanshika Yadav
Killi Akshith Kumar
Saumya Soni
Shrijan Sanidhya
Syed Darain Qamar
B Mohith Venkata Sai Krishna

CONTEXT AND PROBLEM STATEMENT

SECTOR CONTEXT

The aviation industry operates on thin, volatile margins, with jet fuel accounting for about 30% of total costs.

Fuel prices fluctuate due to **global conflicts, economic conditions, and travel demand,** and even small changes can significantly impact airline profits.

Key decision-makers:

- **CEOs & CFOs** → Manage costs and profitability
- **Revenue teams** → Set prices based on demand
- **Pricing teams** → Maintain competitive pricing

PROBLEM STATEMENT

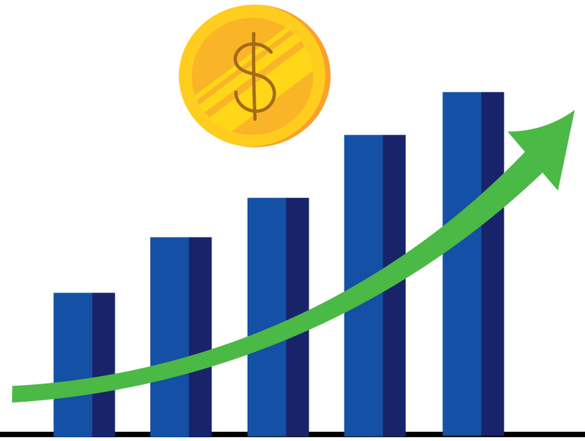
When fuel prices rise, airlines must either absorb costs (hurting profits) or raise ticket prices (risking lower demand). The real challenge is finding the right balance between profitability and demand.

Core Question:
How much of the fuel cost increase is passed on to passengers through ticket prices?
And what other factors influence this decision?

OBJECTIVE

The main goals of this analysis are:

- Measure how much fuel cost affects ticket prices (price pass-through effect)
- Understand other influencing factors, such as:
  - Passenger demand
  - Flight routes
  - Geopolitical events



DATA ENGINEERING (SOURCE → PIPELINE)

Source

- Dataset: Airline Ticket Prices vs Fuel Costs (Kaggle)
- Size: Multi-dataset (~6 datasets merged)
- Time Period: Covers pre-COVID, COVID, and post-COVID phases

Python ETL (Cleaning Steps)

Handled missing values, standardized data, and flagged (not removed) outliers using IQR to retain real-world events like COVID and geopolitical shocks (is_extreme_fare)

GitHub (Pipeline Work)

- **Notebook 01:** Data extraction & cleaning of 6 raw datasets
- **Notebook 02:** Feature engineering (ratios, YoY changes, fare metrics)
- **Final Output:** Unified dataset by merging ticket, fuel, and surcharge data

Data Dictionary (Key Columns)

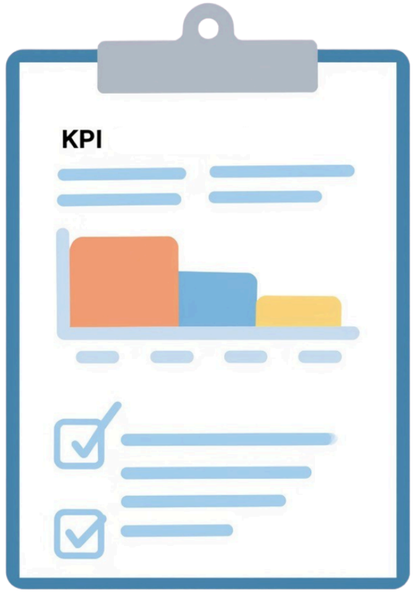
total_fare_usd	**Final ticket price paid by passengers**
jet_fuel_usd_barrel	**Jet fuel price (main cost driver)**
fuel_surcharge_usd	**Additional charge linked to fuel cost**
load_factor_pct	**Seat occupancy (demand indicator)**
conflict_phase	**Market phase (COVID, recovery, conflict, etc.)**

KPI & METRICS FRAMEWORK

Avg Ticket Price & YoY Change	Load Factor	Extreme Fare Records	Average Fuel Surcharge
$31.46	**70.32%**	**287**	**$111.45**

Why these KPIs?

- Directly aligned with the core problem: fuel cost vs ticket pricing
- Help analyze whether airlines absorb costs or pass them strategically
- Capture impact of demand, pricing behavior, and global disruptions



- Average Base Fare
- Average Taxes & Fees
- Average Fare per KM
- Average Fuel Cost % of OpEx

How are they computed?

- Calculated using Python on the cleaned dataset
- Key computations:
  - YoY Change → avg of **yoy_price_change_pct**
  - Load Factor → avg of **load_factor_pct**
  - Extreme Fares → count using **is_extreme_fare flag**
  - Fuel Surcharge → avg of **fuel_surcharge_usd**
- Implemented in the analysis notebook pipeline

KEY INSIGHTS (EDA)

Fuel costs increase airline expenses, but are not fully passed to customers

Ticket prices rise with fuel prices, but not proportionally — indicating strategic pricing rather than direct cost pass-through

Demand and global events drive pricing more than fuel costs

During COVID, prices dropped sharply despite cost factors, while recovery and conflict phases saw strong price surges driven by demand and market conditions

Airlines adjust fuel surcharges instead of base fares

Base fare remains relatively stable, while fuel surcharge increases dynamically during high fuel price periods, showing flexible pricing strategy

Pricing spikes during geopolitical conflicts indicate high market volatility

Conflict phases (e.g., war periods) show higher fares and wider price variation, highlighting the impact of global instability

Long-haul routes are more sensitive to fuel price changes

Flights with longer distances show higher surcharge ratios and greater price fluctuations, making them more vulnerable to fuel shocks

Airline pricing is not purely cost-driven — it is a combination of fuel costs, demand, competition, and global events.

ADVANCED ANALYSIS

What advanced analysis was performed?

Regression (Root Cause Analysis)

- Identified impact of fuel price on ticket price
- Result: Positive relationship, but low explanatory power ($R^2 \sim 0.08$)
- Insight: Pricing is influenced by multiple factors, not just fuel cost

Hypothesis Testing (t-test)

- Compared ticket prices across different phases
- Result: Significant difference between pre-pandemic and conflict periods ($p < 0.05$)
- Insight: Global events strongly impact pricing decisions

Forecasting (ARIMA Model)

- Predicted future ticket prices based on historical trends
- Result: Prices expected to remain high but stable
- Insight: Sudden spikes are event-driven, not trend-driven

Segmentation (K-Means Clustering)

- Grouped data into pricing patterns
- Identified clusters:
  - Low-price (low demand & fuel)
  - Mid-range
  - High-price (high demand & fuel)
- Insight: Different market conditions follow distinct pricing strategies

Tools & Libraries Used

Pandas

NumPy

seaborn

Matplotlib

statsmodels

SciPy

scikit learn contrib

TABLEAU DASHBOARD WALKTHROUGH

Executive View (For Decision Makers)

Provides a high-level overview of business performance

- Key KPIs: Avg Ticket Price, YoY Change, Load Factor, Extreme Fares
- Trend analysis across conflict phases (COVID, recovery, conflicts)
- Region-wise and airline-type comparison for strategic insights

Helps CEOs/CFOs quickly understand pricing trends, demand, and market volatility

Operational View (For Analysts)

Enables deep-dive analysis and decision support

- Fare breakdown: Base Fare, Fuel Surcharge, Taxes & Fees
- Route-level insights: Fare per KM, route class comparison
- Airline-level performance and pricing distribution
- Fuel impact analysis: Jet fuel vs surcharge, cost % of OpEx

Helps analysts understand pricing components and cost behavior in detail

Dashboard Access

https://public.tableau.com/app/profile/syed.darain.qamar7769/viz/AirlinePricesAnalysis/ExecutiveOverview?publish=yes

Interactivity (Live Filtering)

Conflict Phase

Airline Type

Route Class

Year

RECOMMENDATIONS

1. Pricing is not directly driven by fuel cost

Insight: Weak relationship between fuel price and ticket price
Recommendation: Implement advanced dynamic pricing models using demand, seasonality, and route-level factors
Expected Impact: Improved revenue optimization and better pricing competitiveness

2. Fuel cost significantly impacts airline expenses

Insight: Strong impact of fuel prices on operating costs
Recommendation: Adopt fuel hedging strategies and fuel efficiency optimization
Expected Impact: Reduced cost volatility and improved profit margins

3. Long-haul routes are more fuel-sensitive

Insight: Higher fare volatility and surcharge dependency in long-haul routes
Recommendation: Apply route-specific pricing strategies (long-haul vs short-haul)
Expected Impact: Better cost recovery and improved route profitability

4. Pricing spikes during conflicts and crisis periods

Insight: Significant fare increase and volatility during geopolitical events
Recommendation: Build a crisis-responsive pricing framework for real-time adjustments
Expected Impact: Maximized revenue during high-demand or disruption periods

5. Demand plays a key role in pricing decisions

Insight: Load factor influences pricing but is underutilized
Recommendation: Align capacity planning with demand forecasting
Expected Impact: Higher occupancy rates and improved revenue per flight

IMPACT & VALUE

Business Impact (So What?)

Revenue Optimization (+5–10%)
- Dynamic pricing aligned with demand & route behavior
- Reduces underpricing during high-demand periods

Cost Stability (↓ 10–15% volatility)
- Fuel hedging & efficiency strategies reduce exposure to fuel price shocks
- Improves financial predictability

Route Profitability (+8–12%)
- Differentiated pricing for long-haul vs short-haul routes
- Better recovery of fuel-sensitive costs

Operational Efficiency (+3–5% load factor improvement)
- Demand-based capacity planning
- Higher seat utilization → better revenue per flight

Crisis Revenue Gains (+5–8%)
- Faster response to geopolitical events & demand spikes
- Captures high-value pricing opportunities

Cost of Inaction

- **Revenue loss** due to inefficient pricing strategies
- **Reduced margins** from uncontrolled fuel cost impact
- **Missed opportunities** during high-demand or crisis periods
- **Poor demand alignment** → lower load factors & wasted capacity

Why stakeholders should act

The aviation industry is highly volatile — adopting data-driven pricing and risk strategies is essential to remain competitive, profitable, and resilient.

LIMITATIONS & NEXT STEPS

Limitations

Limited explanatory power of model:
Regression shows low R^2 (~0.08), meaning pricing depends on many unseen factors

Missing key variables
Data does not include competition, seasonality, or route-specific demand

External shocks not fully captured
Forecasting models fail to predict sudden events like COVID or geopolitical conflicts

Aggregated data limitation
Analysis is at a high level, not airline-specific or route-specific

Future Scope

Build advanced predictive models
Use SARIMA / ML models with external factors for better accuracy

Incorporate richer datasets
Add demand data, competitor pricing, macroeconomic indicators, fuel hedging data

Apply advanced segmentation
Use clustering to identify airline, route, and region-specific pricing strategies

Use causal analysis techniques
Move beyond correlation to understand true drivers of pricing changes

TEAM & CONTRIBUTIONS

Vanshika Yadav (Project Lead, PPT & Quality)

Led project, handled reporting, and performed data cleaning & statistical analysis.

Shrijan Sanidhya (Data Lead)

Managed dataset sourcing and initial data structuring

Killi Akshith Kumar (ETL Lead)

Built data pipeline, performed cleaning, preprocessing and Conducted EDA

Saumya Soni (Analysis Lead)

Statistical analysis, and insight generation

Syed Darain Qamar (Visualization Lead)

Designed and developed Tableau dashboard

B Mohith Venkata Sai Krishna (Strategy Lead)

Worked on business insights, recommendations, and impact analysis